

EU Code - EU Title

Cycle: **Engineer**

Option/Sector: **Computer Science and Information Systems**

Constituent Element (CE) Code: **ING4178**

EC title : **ISI Transversal project**

Head of the EC: **Arthur PESSA** / Tel: (+237) 675 83 59 53 / Email: arthur.pessa@saintjeaningenieur.org

Other EC speaker(s): **Emmanuel MOUPOJOU** / Tel: (+237) - / Email: emmanuel.moupojou@saintjeaningenieur.com

PROJECT DESCRIPTION

Platform for Connecting Agricultural Cooperatives and Farmers

1. Context and objectives of the project

In Africa, the majority of farmers work in rural areas, sometimes without stable internet access and often with low literacy levels. Agricultural cooperatives play a key role in organizing, pooling resources, and marketing products. The government wants to establish a **digital system for monitoring, controlling, and facilitating trade** .

Goals :

- Create a **centralized digital platform** enabling:
 - Connecting **farmers and agricultural cooperatives** .
 - **Monitoring and control of agricultural activities** by the government.
 - Farmers' access to information even **without an Internet connection** .
- **simple and accessible** interfaces (icons, audio, local multilingual).
- Ensure a **modern and scalable architecture** based on microservices .

2. Target users

1. Farmers :

- Low literacy → simple interface (icons , audio, local languages).
- Priority offline access.
- Each farmer accesses their account through a QR Code or registration number.
- Notifications (market price, weather, cooperative, status of an agricultural project): includes notifications when the cooperative validates a payment.
- Monitoring of debts/microcredits (transparency).

2. Agricultural cooperatives :

- Management of members (farmers).
- Monitoring of agricultural production and stocks.
- Publication of announcements and needs via configurable audio messages in various languages.
- Monitoring of plant health status production (detect if there is any plant disease and provide a report to the farmer (according to his agricultural area)
- Management of microcredits granted and monitoring of repayments.

3. Government :

- Access to agricultural statistics by region.
- Global Deliveries Dashboard by region and production.
- Data export (Excel/PDF).
- Detection and Notification of low production areas (alerts).

3. Main Features

Mobile Application (Farmers)

- Registration via cooperative (QR code or simple number).
- Consultation of market prices.
- Localized weather information.
- Voicemail (cooperative announcements → audio).
- Offline consultation with subsequent synchronization.
- Notifications (market price, weather, cooperative, status of an agricultural project): includes notifications when the cooperative validates a payment.
- Monitoring of debts/microcredits (transparency).

Web application

Government Space:

- Statistics dashboard (by region, crop, cooperative) following a specific period
- Global Deliveries Dashboard.
- Data export (Excel/PDF).
- Alert system (disaster, low performance, diseases)

Cooperative Space:

- Management of member farmers.
- Inventory management and monitoring of agricultural production related of farmers.
- Publication of needs (seeds, fertilizers, equipment).
- Connection for the sale of products.
- Manage microcredits allowed to farmers. Check those who are eligible to bank credits

4. Constraints

- Operation in areas **with low Internet coverage** .
- Accessibility (intuitive interface, audio, multilingual).
- Compliance with security standards (GDPR/Africa Cloud & Local Data).
- **Android** compatibility ≥ 6.0 (low-end phones).

Deliverables

1. **Web application** (responsive interface, progressive web app possible):
 - **Government Side** : Monitoring, control, reporting , macro indicators.
 - **Cooperative Side** : Member management, stocks (delivered production), transactions, contracts, microcredits.
2. **Mobile Application** :
 - Interface adapted to farmers (icons + voice messages).
 - **Offline-first** operation (data stored locally, synchronization when the internet returns).
 - Access to price information , weather, training, cooperative announcements.
3. **REST APIs** exposed through a **microservices architecture** .
4. **Centralized and replicated database** (MySQL).
5. **Complete documentation** :
 - Technical documentation (API, microservices architecture).
 - User manual (cooperative, farmers, government).

System Administration Guide.